

Fairness and reward in Energy Communities: game-theory versus simplified approaches

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Abstract—The energy transition requires huge investments. However, increasing citizen participation in the energy system by providing flexibility and decentralized generation and storage could provide relief and reduce the total system costs. In this regard, several governments are promoting the development of aggregations of users, sometimes in the form of Energy Communities, to foster the desired socio-economic and environmental goals. Energy Communities are expected to attract local investors and induce positive behavioural change that lowers the burden on the power grid. However, business models and fairness in the reward schemes are matters of concern that cannot be disregarded in the optimal planning of Energy Communities. In this framework, this paper proposes a comparison of different schemes that can be used to share revenues and rewards among the members of a Community. The cooperative optimization problem, calculated using Mixed-Integer Linear Programming (MILP), is compared to non-cooperative approaches where users do not cooperate. Simple rewarding schemes, e.g. proportional to the shared energy and investment costs, are compared to game-theory approaches using Shapley Value, Nucleolus, and VarLeast-Core. A numerical case study suggests that the procedure can be implemented in practice and suggests approximate techniques to address reward distribution schemes for Energy Communities.

Index Terms—Aggregator; Energy Service Company (ESCO); microgrid; fairness and efficiency; reward distribution

I. INTRODUCTION

To meet the EU environmental targets, citizen participation in energy decisions is needed. Accordingly, the EU's Clean Energy Package includes legal frameworks for Energy Communities (ECs) [1], [2], which are self-organized legal entities aimed at promoting local energy sharing for social, environmental, and economic benefits. Members of ECs can collectively produce and self-consume renewable energy that is produced by the members of the EC itself, provided that they are awarded an appropriate benefit in exchange. For example, members of ECs can supply their surplus energy or consume surplus energy from their neighbours. Appropriate incentives are needed to induce users to become members of the EC, install renewable assets for the community and induce a change in appliances used to increase the benefits.

The incentive scheme must not harm other users nor be a barrier to users' participation, as specified by EU regulation

[1]: EU regulation demands users shall be free to participate in ECs [2]. The national legislation of EU countries has adopted the same principles, yet adapting the EU framework to the national context. A common approach, adopted in Italy [3] or UK [4] for instance, is to reward the entity EC for every unit of energy the users share and let the EC decide on how to distribute the reward. It is worth noticing that often neither the national nor the local legislation prescribes the specific reward distribution to be implemented among the users. On the one end, this offers the flexibility of each community on how to allocate the shared funds, hence promoting active debate towards a common solution. On the other hand, complexities in the specific legal form and the documents to prepare to arise.

Incentive schemes are highly debated and no unique strategy is available, although more than 10000 citizen-led initiatives are currently available in EU [5]. Several studies have proposed game-theory approaches, either cooperative [6]–[8] or competitive [9], [10]. Competitive schemes may support the creation of local energy markets, where users compete for larger shares of the reward [9], potentially - yet not necessarily - within local energy markets. On one hand, this approach can promote active participation by citizenship; however, it may require higher communication schemes, induce barriers and/or conflicts in the community and lead to a sub-optimal global solution for the community itself, conversely to cooperative schemes. On the other hand, cooperative schemes can be more inclusive as the users are expected to cooperate towards the global solution for the community itself. Moreover, trading wouldn't be a major concern for typical households, so simplified schemes may be more appropriate.

Efficiency and fairness of the rewarding scheme shall be safeguarded to (a) guarantee that each user is rewarded for the actual contribution to the community and (b) the maximum benefit is achieved [6]. In this regard, the Shapley Value is a well-known measure for profit allocation [11] and is sometimes used as a key indicator for fairness. However, as confirmed in the literature [6], [12], the corresponding profit allocation may not be stable, meaning that a subset of the community may be better off in creating a separate EC, in opposition to the general goals of inclusiveness ECs should achieve [2]. Core or Nucleolus overcome this challenge [6],

yet the Core redistribution is not unique and does not guarantee that each user is strictly better off within the community than in another community, whereas the Nucleolus one is computationally heavy to compute and hardly justifiable for the EC context.

To partially overcome this challenge, the LeastCore measure is proposed to guarantee any participant to be better off within the community [6], yet the specific reward distribution may still be not unique. Conversely, the recent VarLeastCore and ShapleyLeastCore measures have been proposed in [6] to reduce this arbitrariness. VarLeastCore selects the profit distribution that minimizes the disproportion of treatment across the EC members, provided that the solution lies within the LeastCore, which ensures the strict stability of the coalition. Instead, ShapleyLeastCore identifies the reward distribution within the LeastCore that is closest to the Shapley Value, yet all limitations of the computational complexity for Shapley and LeastCore still apply. These formulations are accepted measures of fairness and efficiency, however, they involve complex mathematical formulations that is difficult to explain to the general audience; hence approximations are needed.

As reward distribution schemes based on game theory are complex to calculate and explain, researchers have also proposed simplified approaches to distribute the rewards according to easy-to-understand procedures [13], often based on the quantity of energy shared and/or the user's participation into the initial investment. However, the theoretical soundness of those reward distribution is generally lacking [14].

Moreover, the members of a Community hardly have the technical and legal expertise to handle the corresponding technical and bureaucratic tasks needed to operate the EC itself. Accordingly, external consultants in the form of aggregators or Energy Service Companies (ESCO) could fill the gap at an appropriate fee. However, the business model and the rewarding scheme shall be properly tailored to achieve the global optimum [6]. Accordingly, appropriate technical tools for the optimal design are required, as proven by the increasing number of studies on the optimal design of ECs [6], [13].

In this study, we propose a comparison of different reward schemes based on game-theory and simplified approaches to highlight potential synergies and differences. A Linear Programming model is used to calculate the optimal design of Energy Community, based on a realistic business model. A comparison between cooperative and non-cooperative schemes is also proposed and simulated using a numerical case study for Italy. The model has been developed in a Julia package incorporating the desired features.

II. THE BUSINESS MODEL OF THE ENERGY COMMUNITY

To properly design an EC, we first denote the corresponding business model and organizational structure that are the backbone for the corresponding optimization model, needed for the proposed activity. In agreement to the literature [6] and EU regulation [2], the EC has no-profit goals, as such it is a no-profit entity and all the revenues it may collect are shared across its members, after its obligations are met. The EC is

rewarded an economic benefit by a public authority for every unit of energy is shared across its members. As the EC hasn't adequate technical and managerial skills, an external for-profit company, hereinafter referred to as "aggregator", is paid by the EC to optimally design the energy system and operate the assets. To avoid the *Agency problem*, the aggregator is rewarded proportionally to the benefit the members receive with respect to the base case where no cooperation between the users exist [6]. The shared energy is the energy that is produced in a 15-min time horizon by a member and consumed by another member of the community in the same time frame.

According to the European Union definition, an EC shall be a no-profit legal entity whose members are allowed to share energy to reach social, environmental and economic benefits [2]. Therefore, their main goal cannot be the energy trading with the public market and it cannot be a private for-profit company. However, rarely all technical skills needed for performing the optimal design and operation belong within the community itself. Rather, it is likely that the EC delegates the optimal design and operation to an external Energy Service Company (ESCO) or an aggregator, that for the sake of brevity are hereinafter denoted as "aggregator". However, the aggregator is a for-profit company, therefore it shall remunerated according to specific legal agreements that shall be agreed with the EC. On the other hand, the members of the EC shall agree to participate to the EC and their privacy shall be guaranteed by appropriate rules, which are all stated in the membership contract between each user and the EC. The EC does not own shared assets, all assets are owned by and installed at each household. All costs related to the assets are paid by the household itself. The reward is distributed among the members of the EC and the Aggregator in agreement to the membership and service contracts, respectively. The goal of the EC is to maximize the global Net Present Value (NPV) of the EC [13].

III. REWARD DISTRIBUTION

In this section, we denote different reward distribution schemes to distribute the reward awarded to the EC. It is worth noticing that in the simplified reward schemes, the reward distribution is calculated explicitly and consequently, the NPV for each member is calculated. On the contrary, game-theoretic schemes focus on identifying the NPV for each member and the reward distribution is calculated as a post-processing.

A. Game-theoretic schemes

Let $v(J)$ be the function that denotes the benefit for members J to join the EC with respect to the base case configuration. Moreover, let's denote with NPV_j^{CO} the Net Present Value of member (or aggregator) j when it belongs to the EC, whereas NPV_j^{NC} represents the base case value when the user does not belong to the EC and hence its energy system is optimized without considering the shared energy with its neighbours. Game-theoretic function will focus on identifying the quantity $DNPV_j = NPV_j^{CO} - NPV_j^{NC}$ using $v(J)$ and the reward distribution R_j is calculated as a post-processing using (1), where NPC_j denotes the total annualized costs bore

by the consumer and R_j represents the annualized contribution to the objective function of the reward.

$$NPV_j^{CO} = R_j - NPC_j \quad (1)$$

1) *Shapley Value*: The Shapley Value is calculated using the formula in (2), as discussed in [6], [15]. The NPV for user j in the EC increases by a weighted average that measures the average marginal contribution of user j for every possible combination $\hat{J}$ of ECs may be formed among the members J of the EC, aggregator included. It is worth noticing that the number of calculations is exponentially increasing with the size of the EC, hence the Shapley Value may be hard to calculate.

$$DNPV_j^{SV} = \frac{1}{|J|} \sum_{\hat{J} \subseteq J} \binom{|J| - 1}{|\hat{J}|}^{-1} [v(\hat{J}) - v(\hat{J}/j)] \quad (2)$$

2) *Nucleolus*: The Nucleolus function focuses on iteratively maximizing the robustness of the coalition to join using the formulation in (4), where θ denotes the benefit of the worst coalition to join the EC. The problem in (4) is iteratively executed and in every iteration the worst coalition $\hat{J}$ of every iteration is calculated alongside its corresponding value $\theta_{\hat{J}}$. For more details, please refer to [16]. Similarly to the Shapley value, also the Nucleolus is computationally hard to solve.

$$\begin{aligned} \max \theta \quad (3) \\ \sum_{j \in \hat{J}} DNPV_j^{Nuc} - v(\hat{J}) \geq \theta \quad \forall \hat{J} \notin \Gamma \\ \sum_{j \in \hat{J}} DNPV_j^{Nuc} - v(\hat{J}) \geq \theta_J \quad \forall \hat{J} \in \Gamma \end{aligned} \quad (4)$$

3) *Least Core*: The Least Core corresponds to identify the set of objective functions that guarantee that any EC created by any subset of the whole community is strictly worse off by a quantity θ that is the largest possible. In mathematical terms, the value θ and the set are obtained by solving the problem in (5)–(6). It is worth noting that this formulation is a stricter version of the Core, wherein each sub-coalition is not strictly better off within the EC. The mathematical formulation for Core is equivalent to the Least Core but by forcing $\theta = 0$.

$$\begin{aligned} \max \theta \quad (5) \\ \sum_{j \in J} DNPV_j^{LC} - v(J) \geq \theta \quad \forall J \subset I \end{aligned} \quad (6)$$

Least Core can be calculated more efficiently than Shapley and Nucleolus, yet decomposition approaches are needed [17]. However, the Least Core is generally not unique, hence multiple solutions may apply.

4) *Variance Least Core*: To identify a solution within the Least Core, hence avoiding non-repeatable solutions, while keeping fairness and high computational performances, the authors in [6] proposed to introduce to identify the benefit allocation across members of the community to minimize the dis-uniformity of contribution across the users, provided that the solution belongs to the Least Core. Be $\hat{\theta}$ the benefit of the

worst coalition to stay in the EC, the mathematical formulation is described in (7)–(8).

$$\max \sum_{j \in J} (DNPV_j^{VLC} - \frac{1}{|J|} \sum_{j \in J} DNPV_j^{VLC})^2 \quad (7)$$

$$\sum_{j \in J} DNPV_j^{VLC} - v(J) \geq \hat{\theta} \quad \forall J \subset I \quad (8)$$

B. Simplified approaches

Game-theoretic functions may be hard to calculate as they involve the calculation of the marginal contribution of each member in all groups of members. That may be difficult to perform in practice due to computational limitations [6], thus researchers have suggested simplified approaches based on incentives on the shared energy or the initial investment [13].

1) *Sharing energy schemes*: In energy sharing schemes, the total reward awarded to the community may be shared proportionally to the contribution of each user to the shared energy, be it produced E^{SH+} or consumed E^{SH-} , using two equivalent reward tariffs, π^{SH+} and π^{SH-} . Accordingly, the objective function of each user can be represented as in (9)–(10), where constraint (10) guarantees that the total reward equals the total available reward awarded to the community.

$$R_j^{SE} = \pi^{SH+} E^{SH+} + \pi^{SH-} E^{SH-} \quad (9)$$

$$\sum_j R_j^{SE} = \sum_j NPV_j^{CO} - NPC_j \quad (10)$$

This reward scheme is simple and promotes the coordination of consumption and production, provided that adequate rewarding tariffs are selected. However, the reward is not known a-priori, as the shared energy can only be calculated as a consequence of the simulation. Hence, investors in energy assets are subject to investment risks, similarly to game-theoretic schemes. On the other hand, the risks on the EC or aggregator operating the system may be lower because rewards are calculated a-posteriori and no hedging is performed. However, the reward distribution is not backed by strong mathematical background and does not meet the game-theoretic principles of the methodologies discussed in Section III-A, hence the global EC optimum may not be achieved and competitive behaviours may occur.

To simplify the formulation, in the following, we assume that $\pi^{SH+} = \pi^{SH-} = \pi^{SH}$.

2) *Financing schemes*: To reduce the investment risk for investors in renewable generation, alternative schemes rewarding the installation of assets, e.g. renewable assets and/or batteries, have also been proposed [13]. Their mathematical formulation can be represented using (11)–(12), where γ_i^F denotes the reward contribution for every assets i installed by user j and $x_{i,j}$ denotes the quantity of assets installed. γ_0 represents a fixed reward awarded to each user independently

on its contributions into assets. Similarly to the energy sharing scheme, the EC total reward shall be guaranteed with (12).

$$R_j^F = \gamma_0^F + \sum_i \gamma_i^F x_{i,j} \quad (11)$$

$$\sum_j R_j^F = \sum_j NPV_j^{CO} - NPC_j \quad (12)$$

This rewarding scheme has the advantage to be extremely simple to understand and risk-reducing for investors in renewable assets. However, the risks are passed on the aggregator that has to calibrate the coefficients γ_i^F and potential caps on the total installed assets to make sure that the available reward collected during the operation of the system covers the expected tariffs promised to the users. In practice, a mixture of financing and energy sharing schemes may also be applied.

In this study, as a simplification, we assume as a rewarding scheme a quantity strictly proportional to the investment, shown in (13).

$$R_j^F = \gamma^F CAPEX_j \quad (13)$$

IV. THE OPTIMIZATION PROBLEM

A. The objective function

In this subsection, the mathematical modelling of the Cooperative (CO) and Non-Cooperative (NC) problems is depicted with a nomenclature consistent with the reward distributions detailed in Section III.

1) *Non-Cooperative (NC) problem:* The base case problem corresponds to the optimization of each user independently, hence it corresponds to the maximization of the user's Net Present Value, as detailed in (14)–(21), where NPC_j denotes the formulation of the Net Present Cost of the user system, $I_{j,y}^U$ is its investment costs, $RC_{j,y}^U$ highlights the replacement charges, $RV_{j,y}^U$ is the recovery value and $R_{j,y}^P$ denotes the net revenue with respect to the grid, as detailed in (16). Coefficients $\pi_t^{P+/-}$ denote the selling (+) and purchasing (-) prices of electricity and c_w^{PP} represents the peak tariff. $c_j^{a,U}$ is the per-unit investment cost for every unit of installed capacity of the assets and $c_j^{a,M}$ is the maintenance costs. d is the discount rate.

$$NPV_j^{NC} = \max -NPC_j \quad (14)$$

$$NPC_j = \sum_{y \in Y} \frac{C_{j,y}^P + I_{j,y}^U + RC_{j,y}^U - RV_{j,y}^U - R_{j,y}^P}{(1+d)^y} \quad (15)$$

$$R_{j,y}^P = \Delta \sum_{t \in T} m_t^T \left[\pi_t^{P+} P_{j,t}^{U+} - \pi_t^{P-,V} P_{j,t}^{U-} - \pi_t^{P-,F} P_{j,t}^L \right] \quad (16)$$

$$C_{j,y}^P = \sum_{w \in W} m_w^W c_w^{PP} P_{j,w}^{max} \quad (17)$$

$$I_{j,0}^U = \sum_{a \in A_j} x_j^{a,U} c_j^{a,U} \quad (18)$$

$$C_{j,y}^M = \sum_{a \in A_j} x_j^{a,U} c_j^{a,M} \quad (19)$$

$$RC_{j,y}^U = \sum_{a \in A_j} \begin{cases} x_j^{a,U} c_j^{a,U} & \text{mod}(y, N_j^{Y,a}) = 0 \\ 0 & \text{else} \end{cases} \quad (20)$$

$$RV_{j,y}^U = x_j^{a,U} c_j^{a,U} \frac{N_j^{Y,a} - \text{mod}(|Y| - 1, N_j^{Y,a})}{N_j^{Y,a}} \quad (21)$$

2) *Cooperative (CO) problem:* Conversely, the Cooperative (CO) optimization enabled by the aggregator in an EC maximizes the overall Net Present Value NPV^{CO} of the entire community as detailed in the following. It is worth noticing that NPV^{CO} is formulated as the sum of the NPC_j of each member and the term R^{SH} that denotes the reward awarded to the community and defined in (23) with a feed-in-tariff approach.

$$NPV^{CO} = \max R^{SH} + \sum_j -NPC_j \quad (22)$$

$$R^{SH} = \sum_y \frac{\sum_t m_t^T \pi_t^{SH} E_t^{SH}}{(1+d)^t} \quad (23)$$

In the game-theoretic reward distribution depicted in Section III-A, the function v is calculated as the difference between the CO and NC objective functions defined in (22) and (14).

B. Constraints

The technical constraints of the users' systems are described in (24)–(29). Equation (24) guarantees the users' electric balance, where $P_{j,t}^{U+/-}$ denotes the energy fed (+) and purchased (-) from the grid of each user j , $P_{j,t}^{c+/-,U}$ is the power of the converter, $P_{j,t}^{R,U}$ represents the total renewable production and $P_{j,t}^L$ is the load. The peak power of each user is guaranteed with (25). The availability of renewable production is modelled using (26), where $p_{j,t}^{r,U}$ denotes the per-unit renewable production of the renewable asset r . The battery modelling is detailed in (27) and (28); η_j^b is the round-trip efficiency of the battery system b and $c(b)$ denotes the converter c corresponding to the battery b . $x_j^{i,U}$ represents the variable of the investment capacity for the asset i (b : battery, r : renewable asset, c : converter) and user j .

$$P_{j,t}^{U+} - P_{j,t}^{U-} + \sum_{c \in A_j^C} \left[P_{j,t}^{c-,U} - P_{j,t}^{c+,U} \right] - P_{j,t}^{R,U} = -P_{j,t}^L \quad (24)$$

$$P_{j,w}^{max} \geq \left| P_{j,\hat{t}}^{U+} - P_{j,\hat{t}}^{U-} \right| \quad (25)$$

$$P_{j,t}^{R,U} \leq \sum_{r \in A_j^R} p_{j,t}^{r,U} x_j^{r,U} \quad (26)$$

$$E_{j,t}^{b,U} = E_{j,t-1}^{b,U} - \Delta P_{j,t}^{c(b)+,U} / \sqrt{\eta_j^b} + \Delta P_{j,t}^{c(b)-,U} \sqrt{\eta_j^b} \quad (27)$$

$$P_{j,t}^{c+,U} + P_{j,t}^{c-,U} \leq x_j^{c,U} \quad (28)$$

$$x_j^{b,U} \beta_j^{b,min} \leq E_{j,t}^{b,U} \leq x_j^{b,U} \beta_j^{b,max} \quad (29)$$

In the cooperative (CO) formulation, hence when the EC is formed, the shared energy is computed as the minimum between the energy conjointly fed and consumed within the EC as denoted in (30).

$$E_t^{SH} \leq \min \left\{ \sum_{j \in J} \frac{P_{j,t}^{M-}}{\Delta}, \sum_{j \in J} \frac{P_{j,t}^{M+}}{\Delta} \right\} \quad (30)$$

Given the above-mentioned modelling, the function $v(J)$ is calculated in (31) as the difference between the CO and NC configuration for a given set J of members of the community.

$$v(J) = NPV^{CO}(J) - NPV^{NC}(J) \quad (31)$$

V. CASE STUDY

A. Description

The case study consists of a possible Energy Community in Italy composed by commercial users, similarly to [6]. The time series of demand have been derived from the data collected from a Portuguese substation with similar characteristics to the Italian context [18]. Solar and wind have been considered as suitable generating assets and storage; the time series have been derived from the Renewable.ninja platform [19]. All but three users can install photovoltaic assets, two can install also wind turbines and five can install battery assets. The space availability for generating assets and the load demand are based on the input sets stated in [6].

B. Main parameters

The investment costs of PV are in the range 1400-1700€/kWp and space limitations in the range 50-100kWp, depending on the user; the investment costs for wind turbines are 3000€/kWp. Lithium batteries can be installed at a cost of 400€/kWh and 200€/kW (converter), with comprehensive round-trip efficiency of 92%. Lifetime is 25 years for PV assets, 20 for wind turbines, 15 for batteries and 10 for converters. Maintenance charges are in the range 1-2% the investment capacity. Pre-crisis electricity prices have been considered to compensate for recent price fluctuations. Accordingly, purchase and selling prices are 18c€/kWh and 5c€/kWh, respectively, with yearly peak power charges of 36€/kW [3].

C. Test procedure

To compare the reward and cost distribution across the participants, we perform the following procedure:

- 1) Execute the CO and NC optimizations to identify the objective functions and energy flows of the community, including shared energy (consumed and generated) and the design variables
- 2) Execute the game-theoretic reward distribution functions and calculate the reward allocation
- 3) Simulate the use of simplified reward allocation schemes to capture the reward allocation described by a game-theoretic approach for every user independently

Results are obtained and shown in the following section.

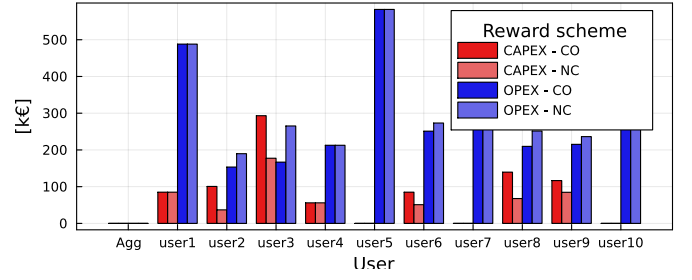


Fig. 1: CAPEX and total discounted OPEX in CO and NC configurations.

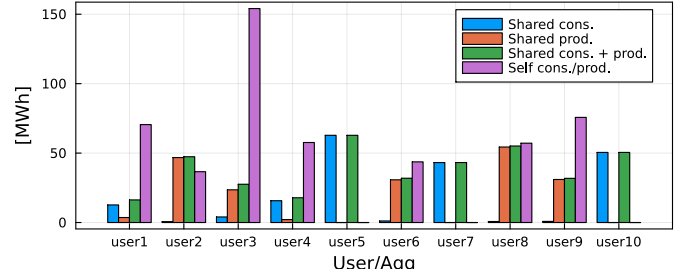


Fig. 2: Shared consumption, shared production and self consumption for the CO configuration.

VI. RESULTS

Major results are shown in Figures from 1 to 5. Fig. 1 highlights the life-cycle discounted CAPEX and OPEX for the CO and NC configurations, as detailed in Section IV. In the CO configuration, increased investments in renewable and battery assets of users 2, 3, 6, 8, and 9 are partially compensated by lower operational costs, the remainder is covered by appropriate rewarding schemes. Indeed, that is enabled by the revenues related to increased levels of shared energy, be the fraction of energy that a user produce but consumed by another user, and also self-consumed, be it the fraction of energy that an user produces and consumes itself, shown in Fig. 2. In the NC configuration no shared energy, be it produced or consumed, is possible, whereas self-consumption is possible but in smaller quantities than in CO.

When game-theoretic approaches are applied, such as Nucleolus, Shapely and VarLeastCore approaches, the total net benefit in terms of DNPV by user is shown in Fig. 3. The picture highlights that Shapely and Nucleolus tend to behave similarly, whereas the VarLeastCore tends to reduce the benefit awarded to the aggregator, in favor of the users. The reward allocation of the different game-theoretic functions is shown in Fig. 4. It is worth noticing that the benefit and reward allocation maps do not match. In particular, prosumers 2, 3 and 8, do receive a large share of reward, but their total benefit is lower. Indeed, a large portion of the reward they receive is to compensate the larger investments needed to maximize the benefits for the community. Given its better reward allocation to the users, in the following we focus on VarLeastCore.

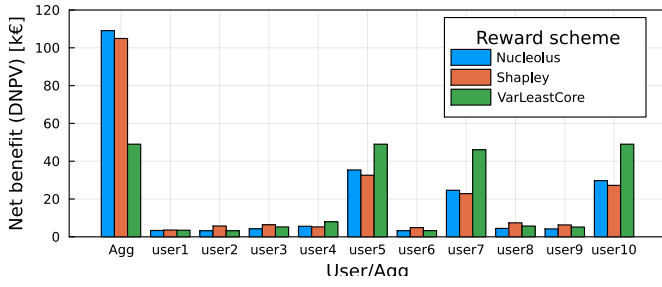


Fig. 3: Benefit in terms of DNPV of CO with respect to NC.

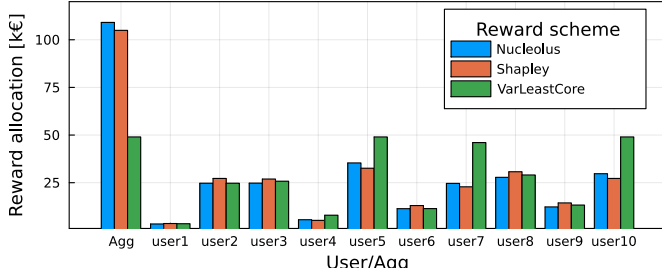


Fig. 4: Reward allocation for game-theoretic functions.

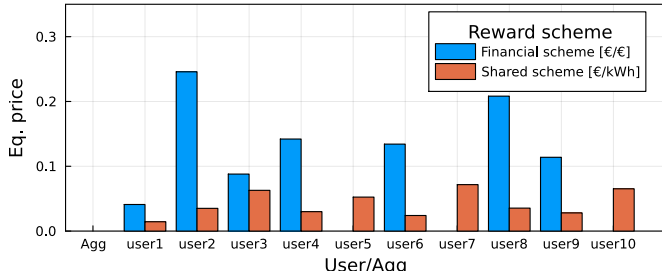


Fig. 5: Equivalent prices for simplified allocation schemes.

Finally, we show in Fig. 5 the equivalent prices for energy sharing scheme (π^{SH}) and for financing schemes (γ^F) personalized by user to match the reward allocation shown in Fig. 4 for VarLeastCore. In agreement to [6], about 16-24% of DNPV is allocated to aggregators as they have the key role of coordinating the users that couldn't otherwise be done by the users alone. It turns out that the equivalent prices are in the range between 1.5 and 7.2 c€/kWh for sharing mechanisms and the financial system values change from 4 to 21c€/€. However, in the case of financial approaches, users 5, 7 and 10, that are pure consumers, require a fixed benefit to assure reward allocation, which introduces uncertainties into the business model of operators. The procedure here described and the numerical considerations can support the business and policy decision making for EC and aggregators.

VII. CONCLUSIONS

This paper has proposed a methodological framework and numerical study to evaluate and compare game-theoretic and simplified schemes aimed at allocating the reward in Energy Communities, subject to feed-in-tariff on shared energy.

Results for a 10-member EC highlight that game-theoretic approach are robust, yet a simplification may be required to implement them in practice. Equivalent incentives for simplified approaches based on feed-in-tariffs on the shared energy show values in the range 1-7c€/kWh, whereas financial CAPEX-based mechanisms are in the range 4 to 21c€/€.

The methodology and the proposed results can lay the foundation for future work on sounded fair efficient reward allocation schemes in business models for ECs.

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